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Unlocking the Sustainable Synergy of Geothermal Energy with ZLD Produced Water Desalination

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Abstract

The conventional thermal based ZLD technologies used for produced water desalination are energy intensive. This study aims to leverage the geothermal energy from abandoned wells to enhance the temperature of pretreated high-salinity produced water and achieve an optimized Zero Liquid Discharge (ZLD) system to significantly lower carbon foot-print associated with produced water desalination. The proposed approach utilizes the closed-loop geothermal heat extracted from the abandoned well to pre-heat the produced water before it enters the evaporation-based thermal desalination unit such as Dynamic Vapor Recovery.

A semi-analytical model, comprising a wellbore heat flow model coupled with a formation heat conduction model, is employed to rapidly determine the thermal energy output potential from the closed-loop geothermal system. The study evaluates the feasibility of increasing the temperature geothermally from 25°C to reach the desired output temperature of 80-85°C by the continuous once through circulation of produced water through an abandoned well. Sensitivity analysis is conducted to study the influence of various parameters, including reservoir characteristics, well parameters, and operational conditions, to identify the optimal scenarios that can result in near zero carbon foot-print ZLD solutions for produced water desalination.

The results of sensitivity analysis indicate that casing size, well depth, flow rate, and reservoir temperature are the most critical parameters affecting the achievable output temperature for the integrated ZLD system. The tubing size was the least influential parameter within the value range (2.5-4.5 inch) studied. It was found that well depths above 10,000 ft and circulation flow rates below 10,000 BPD can reach the desired temperature geothermally with a thermal energy gain of 4.0 Mwth. The reservoir temperatures above 300°F and casing sizes greater than 8.0-8.5 inch are determined to be optimal to provide the sufficient geothermal heat needed for raising the temperature of the pretreated feed produced water into ZLD desalination system without the consumption of any additional heat. Based on these interesting results, it can be concluded that the proposed approach can minimize energy consumption and reduce operational costs, thereby making the geothermal energy integration with produced water desalination both viable and sustainable. In summary, the findings demonstrated the feasibility of leveraging abandoned well geothermal energy to enhance the temperature of high-salinity produced water and significantly lower the energy requirements of ZLD

produced water desalination processes to generate fresh water. These favorable outcomes align very well with UN Sustainability Development Goals 6 and 13 focusing on "clean water", and "climate action", respectively.

The utilization of semi-analytical approach for optimizing the integration of geothermal energy with ZLD technology is a novel contribution to the existing knowledge. This cost-effective synergistic solution can facilitate the broader adoption of sustainable geothermal energy for the desalination of high-salinity produced water, which eventually enhances the environmental and economic viability of ZLD desalination processes to promote both circular water/carbon economies in upstream operations.

Introduction

Geothermal energy is increasingly recognized as a vital renewable energy source due to sustainable energy harvesting, its ubiquitous availability, and consistent accessibility. This energy form harnesses heat from the Earth's interior, which, if effectively exploited, can significantly contribute to reducing dependency on fossil fuels and mitigating environmental impacts (Jiang et al. 2019). The classification of geothermal systems can be broadly divided into open-and closed-loop systems. Open-loop systems, such as conventional hydrothermal reservoirs, allow direct interaction between reservoir fluids and the surface, while enhanced geothermal systems (EGS) utilize artificially created permeable rock formations to facilitate fluid flow (Cheng et al., 2021). Conversely, closed-loop systems circulate a working fluid through a set of pipes, avoiding direct contact with the geothermal reservoir, which can improve environmental sustainability and reduce operational risks (Fan et al., 2019).

This paper studies closed-loop geothermal systems, particularly leveraging the recent in-house developed geothermal screening tools. These tools facilitate the integration of Zero Liquid Discharge (ZLD) water treatment technology, a critical component for sustainable geothermal energy management. ZLD minimizes wastewater generation by recovering and recycling water, thus promoting efficient resource use without harming the environment (Capocelli et al., 2020; Utama et al., 2024). Our focus to achieve ZLD with hypersaline produced water streams includes the thermal based Dynamic Vapor Recovery desalination system, which is designed to operate effectively at specific temperatures, notably around 80-85°C and generates high quality distillate water with up to 80% recovery efficiency. The preparation of fluids for thermal based ZLD desalination systems necessitates heating from typical ambient temperatures of approximately 25 °C to the required operational temperature, a process achievable by utilizing a nearby abandoned geothermal well with a substantial flow rate of 5,000 to 10,000 bbl/day.

The proposed sustainable approach aims to optimize the usage of geothermal resources while simultaneously addressing environmental concerns related to liquid waste management. The incorporation of ZLD in geothermal operations not only aids in adhering to regulatory frameworks demanding sustainability but also enhances the overall efficiency of geothermal systems by reducing losses associated with liquid discharges. Furthermore, implementing the dynamic vapor recovery system within geothermal frameworks can provide substantial economic benefits through energy production and improved heating efficiencies, thus contributing to the broader adoption of geothermal energy in various sectors. ZLD solutions relying on dynamic vapor recovery itself can convert high-salinity produced water of up to over 200,000 ppm into fresh water. Instead of evaporating the water by using external heat in this process, we investigate the potential of utilizing abandoned wells for the necessary thermal input. Figure 1 illustrates the proposed process harnessing the geothermal energy from abandoned wells for desalinating produced water in a ZLD system. The pretreated produced water after the removal of dissolved H₂S and dispersed oil is circulated through the annulus of wellbore to extract the geothermal heat. The heated produced water so obtained from geothermal energy harvesting is then desalinated, without the need of additional energy, using the dynamic vapor recovery system to generate low salinity water. Ultimately, such a zero or low carbon foot-print Zero Liquid Discharge (ZLD) water management solution can solve a major

environmental challenge by using oil field produced water to generate desired injection water with minimal energy requirements for a wide range of upstream applications, such as water re-injection, water for crude oil desalting, water for unconventional fracking, and water for irrigation systems.

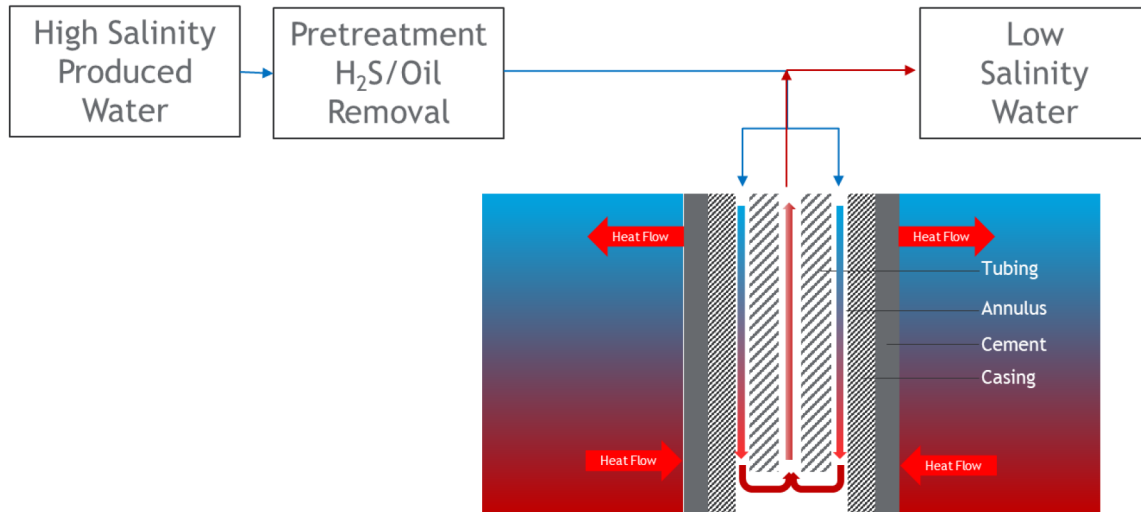


Figure 1—Illustration of the process harnessing geothermal energy from abandoned wells for desalinating produced water in a ZLD system.

Procedure

This study utilizes a semi-analytical model to estimate the heat extraction potential of a coaxial closed-loop geothermal system (CLGS). The modeling approach is strictly based on the formulation presented by Jin et al. (2024). The framework links radial heat conduction in the surrounding formation with a simplified thermal transport model for fluid flow inside the wellbore.

The subsurface environment surrounding the well is treated as a series of horizontal layers, each characterized as homogeneous and isotropic with infinite lateral extension. Because vertical conduction has a minimal impact over the system's depth, the model assumes radial heat transfer to be the dominant mode of thermal exchange. The temperature field within the formation is governed by the transient heat conduction equation in cylindrical coordinates:

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T(r, t)}{\partial r} \right) = \frac{1}{\alpha_f} \frac{\partial T(r, t)}{\partial t}$$

In this equation, $T(r, t)$ represents temperature as a function of radial distance and time, while the thermal diffusivity α_f is given by the relation $\alpha_f = \frac{k_f}{\rho_f c_{p,f}}$, where k_f denotes thermal conductivity, ρ_f is the formation density, and $c_{p,f}$ is its specific heat capacity.

Initial and boundary conditions are specified as follows. At the beginning of production, the formation is uniformly at temperature T_i . A steady heat flux q_w is imposed at the wellbore wall at radius r_w , while the outer boundary at radius r_e is assumed to be thermally insulated. These conditions are mathematically expressed as:

$$\begin{aligned} T(r, 0) &= T_i \\ -2\pi r_w k \left. \frac{\partial T}{\partial r} \right|_{r=r_w} &= q_w \\ \left. \frac{\partial T}{\partial r} \right|_{r=r_e} &= 0 \end{aligned}$$

To facilitate analytical treatment, the model introduces dimensionless variables. The radius, time, and temperature are respectively scaled as $r_D = \frac{r}{r_w}$, $t_D = \frac{4at}{r_w^2}$ and:

$$T_D = \frac{2\pi k_f(T - T_i)}{q_w}$$

Using these dimensionless parameters, the temperature distribution in the formation is described by the exponential integral function:

$$T_D = E_i\left(-\frac{r_D^2}{4t_D}\right)$$

This solution is consistent with established geothermal modeling literature, including foundational work by Ramey (1962) and Hasan and Kabir (1991). The accuracy of this formulation has been verified through comparison with detailed 3D numerical simulations as shown in the referenced study.

The second component of the model focuses on heat transfer within the wellbore, which is configured as a coaxial loop. In this setup, working fluid is injected through the outer annulus (Santos and Taleghani, 2023), flows downward while absorbing heat from the rock formation, and then returns to the surface through the inner tubing as shown in Figure 2. The fluid's enthalpy is defined as:

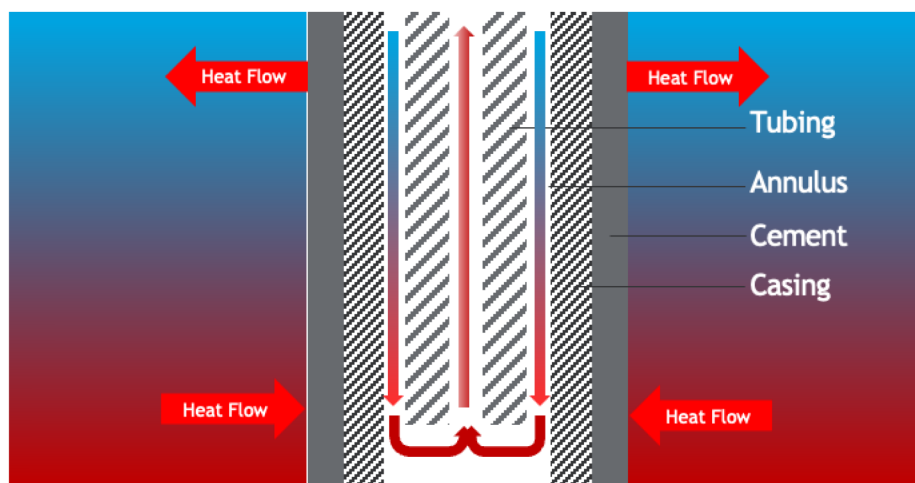


Figure 2—Cross-sectional schematic of a coaxial closed-loop geothermal well. Working fluid descends through the annulus and returns to the surface via the inner tubing. Heat is transferred from the surrounding formation through the casing and cement, primarily into the annular flow, enabling thermal energy extraction at the surface.

$$H = U + pV$$

where H is the total enthalpy, U the internal energy, p the pressure, and V the fluid volume. Heat gained from the inner tubing is labeled dq_{ta} , while heat lost to the formation is represented as dq_{af} .

Although a direct differential expression for temperature variation is not provided, the model solves the thermal interaction between the rock and fluid iteratively using the semi-analytical solution described above in conjunction with the energy balance as shown in Figure 3.

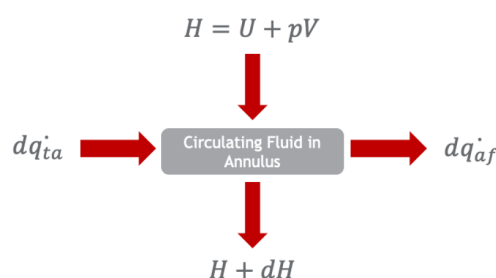


Figure 3—Schematic representation of the energy balance for circulating fluid in the annulus of a closed-loop geothermal system.

The model requires inputs such as thermal conductivity, specific heat, density of the formation, well dimensions, flow rate, initial injection temperature, and operational duration. The outputs include temperature distribution of the circulating fluid and surface heat extraction rate.

Overall, this modeling framework offers a fast and resource-efficient alternative to full numerical simulations. Its ability to produce reliable thermal predictions with relatively minimal data makes it highly suitable for early-stage assessment and feasibility studies, particularly in evaluating the potential of existing oil and gas wells for geothermal redevelopment.

Results and Discussion

To evaluate the viability of integrating geothermal energy into Zero Liquid Discharge (ZLD) desalination systems for high-salinity produced water, a comprehensive sensitivity analysis was performed. Key operational and subsurface parameters were varied to assess their influence on two primary outputs: (1) the temperature of the fluid at the surface, and (2) the thermal energy gain (MWth), both of which are essential for supporting thermal desalination units, such as Dynamic Vapor Recovery without the need for supplemental heating.

Well Depth

Well depth exhibited a strong positive correlation with both surface fluid temperature and thermal energy gain (Figure 4). Deeper wells ($\geq 10,000$ ft) consistently achieved output temperatures exceeding 200°F and thermal energy gains up to 11.87 MWth. This highlights the critical role of geothermal gradient accessibility in achieving high thermal performance and confirms well depth as a primary driver of system efficiency.

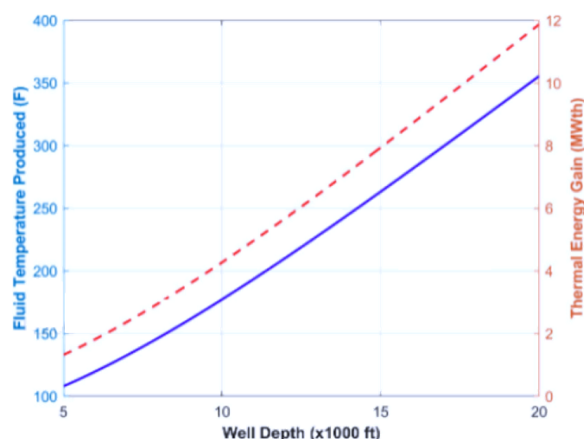


Figure 4—Comparison of working fluid temperature (blue solid line) and thermal energy gain (red dashed line) with respect to well depth in a CLGS.

Flow Rate

An inverse relationship was observed between flow rate and output temperature (Figure 5). As flow rates increased from 5,000 to 15,000 barrels per day (bpd), output temperature decreased from approximately 214 °F to 155 °F due to reduced residence time and heat transfer efficiency. Despite this, total thermal energy gain increased modestly with flow. Therefore, moderate flow rates (5,000-10,000 bpd) are recommended to optimize both thermal efficiency and energy recovery.

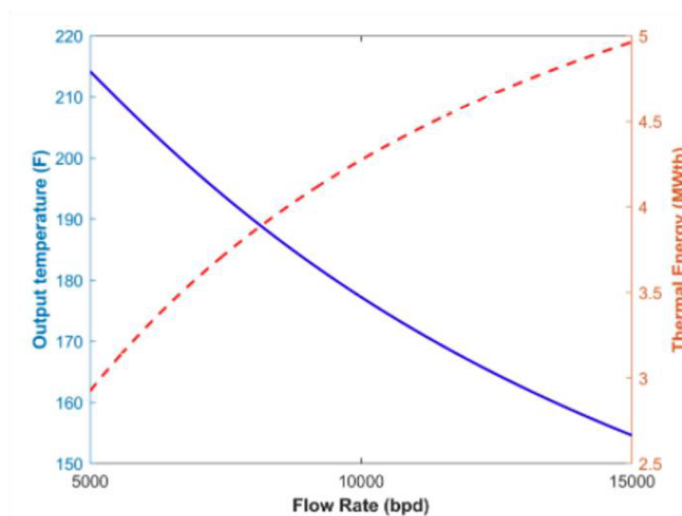


Figure 5—Variation of output fluid temperature (blue solid line) and thermal energy gain (red dashed line) with flow rate in a CLGS.

Injection Fluid Temperature The initial temperature of the injected fluid plays a significant role in determining thermal energy gain (Figure 6). The graph shows that lower injection temperatures result in higher thermal energy recovery, while output temperature increases linearly. This suggests that maintaining a low inlet temperature will maximize the heat extraction efficiency and enables the system to achieve the desired output temperature target range of 80-85 °C.

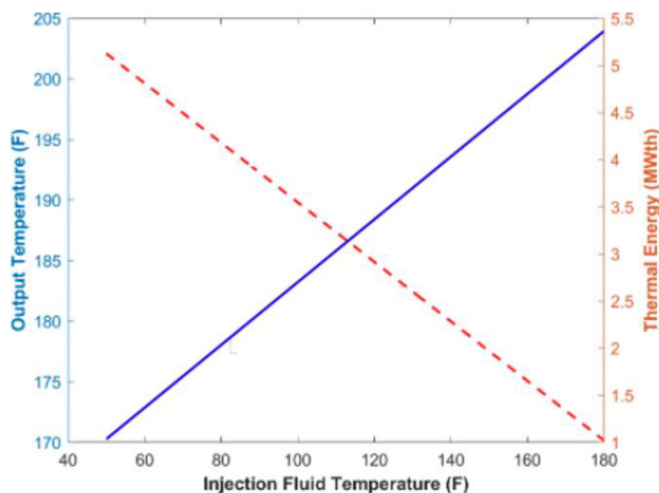


Figure 6—Influence of injection fluid temperature on output temperature (blue solid line) and thermal energy gain (red dashed line) in a CLGS.

Production Duration

The effect of circulation/production time on output fluid temperature and thermal energy gain are shown in Figure 7. Short-term operational durations (1–2 days) resulted in only minor variations in output temperature and energy gain, with a slight decline (approximately 3.7 °F). These findings suggest that the system remains thermally stable during initial operations. However, long-term performance and the potential for thermal breakthrough warrant further investigation.

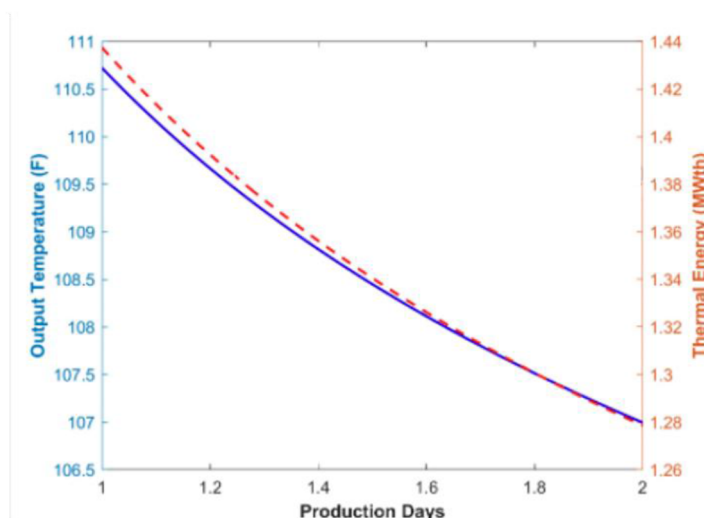


Figure 7—Variation of output fluid temperature (blue solid line) and thermal energy gain (red dashed line) with production time in a CLGS.

Casing and Tubing Sizes

Casing diameter had a notable influence on thermal performance (Figure 8). Increasing casing size from 7 to 10 inches resulted in a substantial rise in surface fluid temperature (130.47 °F to 270.62 °F) and thermal energy gain (2.28 to 8.25 MWth). Conversely, tubing diameter had a negligible impact; across sizes from 2 to 4.5 inches (Figure 9), the variation in both outputs was minimal. This indicates that optimizing casing size is critical for heat transfer, whereas tubing size can be selected based on other operational constraints.

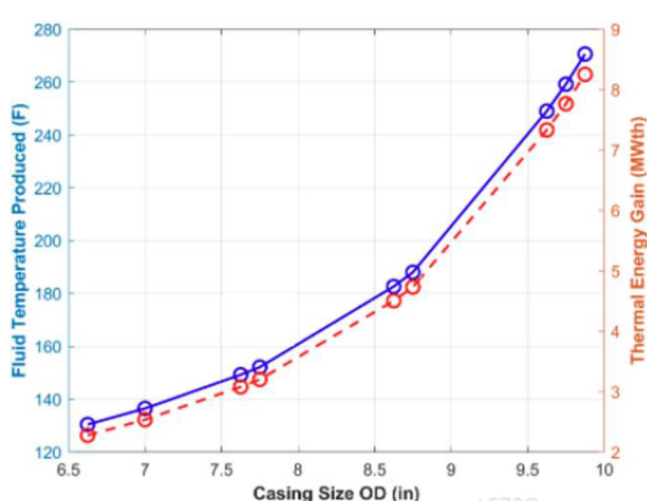


Figure 8—Influence of casing outer diameter on produced fluid temperature (blue solid line) and thermal energy gain (red dashed line) in a CLGS.

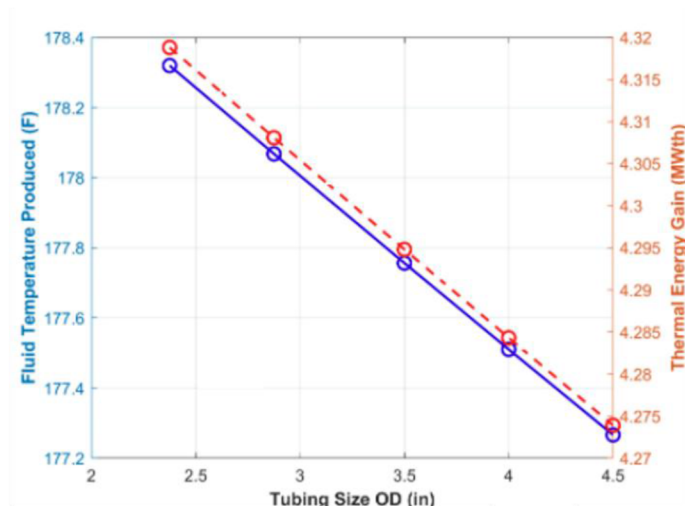


Figure 9—Influence of tubing outer diameter on produced fluid temperature (blue solid line) and thermal energy gain (red dashed line) in a CLGS.

Reservoir Temperature

Reservoir temperature emerged as the most sensitive and influential parameter (Figure 10). Higher reservoir temperatures (≥ 300 °F) yielded output fluid temperatures exceeding 250 °F and thermal energy gains up to 11.94 MWth. A near-linear relationship was observed between reservoir temperature and system outputs, confirming subsurface heat availability as the principal determinant of geothermal integration success.

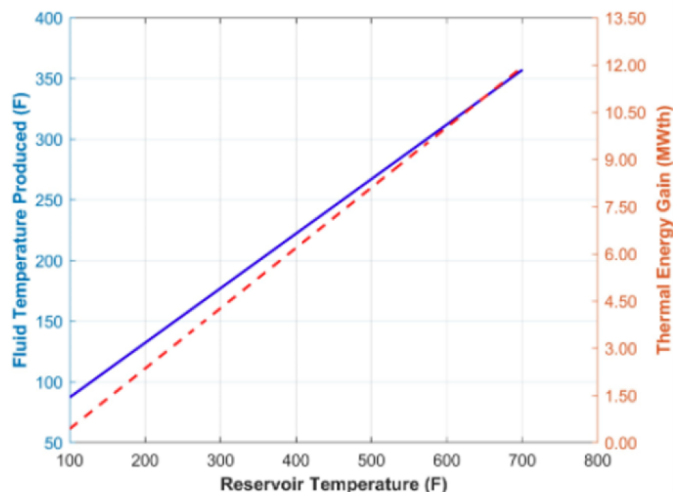


Figure 10—Relationship between reservoir temperature and produced fluid temperature (blue solid line) and thermal energy gain (red dashed line) in a CLGS.

Figure 11 illustrates the temperature profile of circulating fluid within a closed-loop geothermal system as a function of well depth. In this configuration, the working fluid flows downward through the annulus, where it gradually heats up by absorbing thermal energy from the surrounding rock formation. Upon reaching the bottom, the fluid then flows upward through the tubing, largely maintaining its temperature due to insulation and limited thermal exchange. As a result, the annulus fluid shows a variable temperature profile, while the tubing fluid remains nearly constant throughout its upward path.

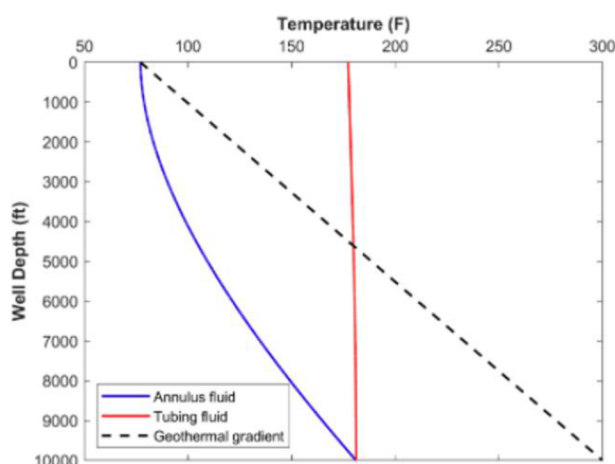


Figure 11—Temperature profiles of annulus fluid (blue), tubing fluid (red), and geothermal gradient (black dashed line) in a closed-loop system. The annulus fluid heats up with depth, while the tubing fluid remains nearly constant during ascent.

The summary of sensitivity analysis results on different parameters such as reservoir characteristics, well specifications, and operational conditions for efficient geothermal energy harvesting that can result in zero/low carbon foot-print ZLD produced water desalination systems is presented in Table 1. As can be seen, the results confirm that well depth, reservoir temperature, casing diameter, and flow rate are the dominant parameters influencing geothermal heat recovery. Such findings have significant implications for designing energy-efficient, geothermal-assisted ZLD systems. While tubing size and short-duration operations have limited thermal impact, strategic optimization of the critical parameters identified can reduce the system's carbon footprint and enhance desalination performance. These insights contribute to the broader goal of achieving sustainable produced water management to generate high quality fresh water with minimal carbon foot-print in the oil and gas sector.

Table 1—Summary of Sensitivity Analysis Results

Parameter	Test Range	Fluid Temp (°F)	Thermal Energy Gain (MWth)	Sensitivity Level	Insight
Well Depth	5,000–20,000 ft	108.01–355.47	1.32–11.87	High	Deeper wells enhance thermal recovery and fluid output temp
Flow Rate	5,000–15,000 bpd	214.19–154.63	2.92–4.96	Moderate	Lower rates favor higher temps; trade-off with energy yield
Injection Temperature	50–180 °F	170.27–203.94	5.13–1.02	High	Lower inlet temp maximizes geothermal heat gain
Production Duration	1–2 days	110.72–106.99	1.44–1.28	Low	Minimal short-term decline; suggests initial system stability
Casing Size	7–10 in	130.47–270.62	2.28–8.25	High	Larger casings significantly boost heat transfer
Tubing Size	2.5–4.5 in	178.32–177.27	4.32–4.27	Very Low	Tubing size has negligible thermal impact
Reservoir Temperature	100–700 °F	87.34–357.12	0.44–11.94	Very High	Dominant parameter governing geothermal potential

Conclusions

This work introduces an innovative, energy-efficient solution for produced water desalination by utilizing geothermal energy from abandoned oil and gas wells to power Zero Liquid Discharge (ZLD) systems. Traditional thermal ZLD technologies, while effective, are highly energy-intensive and contribute significantly to operational costs and carbon emissions. By employing a closed-loop geothermal system to preheat high-salinity produced water before it enters thermal desalination units such as Dynamic Vapor Recovery, this study demonstrates a sustainable and low-carbon alternative.

A semi-analytical model combining wellbore heat transfer with formation heat conduction was developed to evaluate the thermal output potential of abandoned wells. The model assessed the feasibility of heating the injected produced water from an initial temperature of 25 °C to the target range of 80–85 °C using natural geothermal gradients. The results indicate that this temperature increase is achievable under favorable geological and operational conditions, validating the viability of geothermal preheating for thermal based ZLD produced water desalination processes.

Comprehensive sensitivity analysis was conducted to determine how key variables affect system performance, particularly the final fluid temperature and thermal energy gain. The analysis identified well depth, reservoir temperature, flow rate, and casing size as the most influential parameters. Deeper wells (exceeding 10,000 ft) and high reservoir temperatures (above 300 °F) significantly enhanced the thermal energy transfer, enabling output temperatures that meet ZLD requirements without auxiliary heating. Lower flow rates, particularly below 10,000 barrels per day, were also shown to improve thermal efficiency by allowing longer residence time in the well, which facilitates greater heat absorption. Larger casing sizes were found to improve heat exchange capacity, while tubing size had negligible impact across the tested range.

The choice to inject water at 25 °C proved optimal, representing a realistic scenario aligned with surface pre-treatment temperatures and allowing maximum thermal energy gain from the geothermal source. This operational strategy supports the goal of reducing external energy input and minimizing the carbon footprint of the entire desalination process.

Overall, the findings highlight the technical and environmental benefits of integrating geothermal energy with produced water desalination systems. By leveraging the latent thermal potential of abandoned wells, the proposed approach offers a cost-effective and sustainable alternative to conventional ZLD technologies heavily relying on non-renewable energy resources. This integration not only addresses the challenge of managing high-salinity wastewater in upstream operations but also contributes meaningfully to global sustainability goals, particularly UN SDG 6 (Clean Water and Sanitation) and SDG 13 (Climate Action). The study demonstrates that repurposing existing infrastructure for renewable applications can be a transformative step toward a circular carbon and water economy in the energy sector.

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